

Pressure Injury Prevention and Repositioning Protocols

Table of Contents & Navigation Map

Section 1: Skin Anatomy & Pathophysiology

- Page 03: Anatomy of the Skin
- Page 04: Pathophysiology of Tissue Death

Section 2: Risk Assessment & Physical Forces

- Page 05: High-Risk Bony Prominences
- Page 06: Clinical Risk Assessment Tools
- Page 07: Mechanics: Friction vs. Shear
- Page 08: Microclimate Management

Section 3: Clinical Support & Staging

- Page 09: Nutrition & Hydration
- Page 10: Injury Staging: Stage 1
- Page 11: Injury Staging: Stage 2
- Page 12: Injury Staging: Stage 3
- Page 13: Injury Staging: Stage 4
- Page 14: Unstageable & Deep Tissue Pressure Injury (DTPI)

Section 4: Repositioning Protocols

- Page 15: Principles of Repositioning
- Page 16: The 30-Degree Lateral Tilt
- Page 17: Seated Repositioning
- Page 18: Support Surfaces (Mattresses)
- Page 19: Heel Offloading Protocols
- Page 20: Bed Mobility & Slide Sheets

Section 5: Specialized Care & Compliance

- Page 21: Skin Cleansing & Barriers
- Page 22: Medical Device-Related Injuries (MDRPI)
- Page 23: Special Care Populations
- Page 24: Clinical Documentation Rules
- Page 25: Final Competency Checklists

Anatomy of the Skin and Vulnerabilities

Understanding the structure of the skin is essential to understanding how pressure injuries develop and deteriorate.

Skin Layer	Anatomical Function	Clinical Vulnerability
Epidermis (Top)	Waterproof barrier, creates skin tone, protects against pathogens.	Thins rapidly with age; highly susceptible to friction tears and moisture maceration.
Dermis (Middle)	Contains tough connective tissue, hair follicles, and sweat glands.	Loses elasticity and collagen over time; shearing tears the blood vessels located here.
Subcutaneous (Bottom)	Made of fat and connective tissue; acts as an anatomical shock absorber.	Fat atrophy in the elderly or malnourished removes the protective padding over bones.

Pathophysiology: How Tissue Dies

Pressure injuries are the result of localized oxygen starvation. The cycle follows a predictable, deadly path:

External Pressure → Capillary Compression → Ischemia → Necrosis

- **Capillary Closing Pressure:** Healthy capillary blood pressure is roughly 32 mmHg. When external pressure (from a bed or chair) exceeds this, the blood vessels are squeezed shut.
- **Ischemia:** The starvation of oxygen and nutrients to the local tissue due to blocked blood flow. Metabolic waste products rapidly build up.
- **Reactive Hyperemia:** When pressure is relieved, blood rushes back to the area, causing a red flush. If the redness disappears when pressed (blanchable), tissue damage was averted.
- **Necrosis:** If the redness does *not* blanch white when pressed, the lack of oxygen has caused irreversible cell death (necrosis). A pressure injury has officially formed.

High-Risk Bony Prominences

A bony prominence is an area where a bone lies close to the skin surface with very little muscle or fat padding. Gravity sandwiches the skin tightly between the bone and the external surface.

- **Supine (Lying on Back):** Occiput (back of head), Scapulae (shoulder blades), Sacrum (tailbone), Heels.
- **Lateral (Lying on Side):** Ears, Acromion process (shoulder edge), Greater Trochanter (outer hip), Lateral/Medial Condyles (inner/outer knees), Malleolus (ankle bone).
- **Seated (In Chair):** Ischial Tuberosities ("sit bones" of the pelvis), Sacrum, Scapulae, Heels.

Clinical Risk Assessment Tools

Facilities mandate standardized assessments to objectively score a patient's risk upon admission and continuously throughout their stay.

Assessment Tool	Key Variables Measured	Interpretation
Braden Scale	Sensory Perception, Moisture, Activity, Mobility, Nutrition, Friction & Shear.	Scores range from 6 to 23. Lower scores mean higher risk. (e.g., Score ≤ 9 indicates Severe Risk).
Waterlow Score	Build/Weight, Continence, Skin Type, Mobility, Sex/Age, Appetite, Medications.	Scores range upwards. Higher scores mean higher risk. (e.g., Score ≥ 20 indicates Very High Risk).

Trigger Protocol: If an assessment flags a patient as high risk, a preventative care plan must be implemented immediately. Do not wait for an injury to appear.

The Mechanics of Friction vs. Shear

Pressure alone does not cause all injuries. Mechanical forces play a massive role in tissue destruction.

- **Friction:** Two surfaces rubbing together (e.g., skin against a bed sheet). Friction removes the top epidermal layer, causing "sheet burns" and superficial breakdown.
- **Shear:** Internal tissue sliding against itself. The outer skin sticks to the bed, while the underlying skeleton slides downward, stretching and tearing deep blood vessels.

The Head-of-Bed Danger:

Elevating the head of the bed higher than 30 degrees creates massive shearing forces on the sacrum as gravity pulls the patient's torso downward.

Prevention: Always use the knee-break function on a hospital bed to secure the pelvis *before* elevating the headrest.

Microclimate Management

The "microclimate" refers to the temperature and humidity level of the skin precisely where it meets the support surface.

- **Maceration:** Over-hydration of the skin caused by prolonged exposure to sweat, urine, or feces. Macerated skin appears white, wrinkled, and incredibly fragile.
- **The Temperature Threat:** For every 1°C increase in skin temperature, the tissue's metabolic demand for oxygen increases by 10%. If capillaries are compressed, the tissue starves and dies much faster.

Core Interventions:

- Change soiled continence aids immediately.
- Avoid double-padding continence aids, which traps immense heat.
- Use breathable, moisture-wicking underpads.

Nutrition and Hydration Interventions

Skin integrity relies heavily on internal building blocks. A malnourished patient cannot maintain healthy skin or heal existing damage, regardless of how often you turn them.

Nutritional Component	Role in Skin Integrity	Clinical Warning Signs of Deficiency
Protein	Essential for tissue synthesis, collagen formation, and cell repair.	Muscle wasting, generalized edema (swelling), delayed wound healing.
Fluids	Maintains blood volume for oxygen transport and skin elasticity.	Poor skin turgor (tenting), dry mucous membranes, concentrated urine.

Vitamin C & Zinc	Crucial for immune function and structural collagen stability.	Bleeding gums, fragile skin, chronic infections.
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Pressure Injury Staging: Stage 1

Non-Blanchable Erythema of Intact Skin

- **Visual Presentation:** Intact skin with a localized area of redness that does **not** blanch (turn white) when gently pressed with a finger.
- **Additional Indicators:** The area may be painful, firmer or softer than surrounding tissue, and warmer or cooler to the touch. In darkly pigmented skin, redness may not be visible; look for a darker, purplish, or bluish hue and rely heavily on changes in skin temperature.
- **Pathology:** Indicates early-stage tissue ischemia. The surface is unbroken, but damage has begun below.
- **Action Required:** Immediate and total offloading of pressure. Document size, location, and color. Notify nursing staff.

Pressure Injury Staging: Stage 2

Partial-Thickness Skin Loss with Exposed Dermis

- **Visual Presentation:** The wound bed is viable, pink or red, and moist. It presents as a shallow open ulcer. It may also present as an intact or ruptured serum-filled blister.
- **What is NOT Present:** Adipose (fat) tissue is not visible. Deeper tissues are not visible. Granulation tissue, slough (yellow dead tissue), and eschar (black dead tissue) are **not** present in Stage 2.
- **Pathology:** The epidermis is broken or lost, and the upper layer of the dermis is exposed. Usually caused by a combination of pressure, friction, and shear over the pelvis or heel.
- **Action Required:** Implement offloading protocol. Requires clinical wound dressing to protect the exposed dermis and maintain a moist healing environment.

Pressure Injury Staging: Stage 3

Full-Thickness Skin Loss

- **Visual Presentation:** Full-thickness loss of skin. Adipose (subcutaneous fat) is visible in the ulcer. Granulation tissue (bumpy, red, healing tissue) and epibole (rolled wound edges) are often present. Slough or eschar may be present but do not obscure the depth of tissue loss.
- **What is NOT Present:** Fascia, muscle, tendon, ligament, cartilage, and bone are **not** exposed.

- **Pathology:** The injury has eaten completely through the epidermis and dermis into the fat layer. Depth varies by anatomical location (e.g., the bridge of the nose has no fat, so a Stage 3 there is very shallow).
- **Action Required:** Strict repositioning regimen. Advanced nursing wound care, possible debridement of necrotic tissue, and potential nutritional supplements.

Pressure Injury Staging: Stage 4

Full-Thickness Skin and Tissue Loss

- **Visual Presentation:** Full-thickness skin and tissue loss with exposed or directly palpable fascia, muscle, tendon, ligament, cartilage, or bone in the ulcer.
- **Additional Indicators:** Slough or eschar may be visible. Epibole (rolled edges), undermining (tunnels beneath the skin edge), and sinus tracts are very common.
- **Pathology:** Massive structural tissue death extending through the subcutaneous layer down to the musculoskeletal architecture.
- **Action Required:** High risk of osteomyelitis (bone infection) and systemic sepsis. Requires intensive medical management, specialized support surfaces, and potentially surgical intervention.

Unstageable & Deep Tissue Pressure Injury (DTPI)

Sometimes the true depth of the injury is hidden from visual assessment.

Injury Type	Visual Presentation	Clinical Meaning
Unstageable	Full-thickness tissue loss covered by slough (yellow/gray) or eschar (brown/black) in the wound bed.	The true depth cannot be determined until enough dead tissue is removed to expose the base. It is a Stage 3 or 4 hidden by debris.
Deep Tissue (DTPI)	Intact or non-intact skin with localized, non-blanchable deep red, maroon, or purple discoloration.	Intense pressure and shear force have severely damaged the muscle-bone interface deep inside. It can rapidly deteriorate into a Stage 3 or 4 ulcer.

Principles of Repositioning

Repositioning temporarily removes external pressure, allowing compressed capillaries to reopen and deliver oxygen (reperfusion).

- **Frequency:** Dictated by risk score. Standard high-risk protocols require turning every 2 hours in bed, and shifting weight every 30-60 minutes in a chair.
- **Posture:** Maintain spinal alignment and avoid 90-degree resting positions which place total body weight on the hip bone.
- **Equipment:** Use slide sheets and lifters. Never drag or pull a patient.
- **Placement Rule:** Never position a patient directly onto an existing pressure injury or a reddened bony prominence.

The 30-Degree Lateral Tilt

Lying flat on the back puts maximum pressure on the sacrum. Lying fully on the side puts maximum pressure on the greater trochanter (hip bone). The 30-degree tilt distributes weight across the fleshy part of the buttocks, avoiding both bony extremes.

Execution Steps:

1. Roll the patient slightly to one side.
2. Place a positioning wedge or pillows tucked firmly under the back from the shoulder to the pelvis to maintain a precise 30° angle.
3. Ensure the lower hip is pulled slightly forward to avoid direct point pressure.
4. Place a pillow between the knees to protect the medial condyles from rubbing together.

Seated Repositioning

Sitting in a chair concentrates almost the entire body weight onto a very small surface area (the ischial tuberosities or "sit bones"). **Patients break down faster in chairs than in beds.**

- **Time Limits:** High-risk patients should not sit in a chair for longer than 2 hours at a single stretch without returning to bed for pressure relief.
- **Weight Shifting:** If a patient is cognitively intact and has upper body strength, teach them to perform small "push-ups" off the chair armrests every 15 minutes.
- **Posture:** Ensure the patient's feet rest flat on the floor or on footrests. Dangling feet pulls the pelvis forward, creating massive shearing forces on the sacrum.

Support Surfaces (Mattresses)

Support surfaces do not cure pressure injuries; they simply increase the surface area to distribute weight more evenly. **A high-tech mattress never replaces the need for manual turning.**

Surface Type	Mechanism of Action	Clinical Application
Reactive / Static (High-Density Foam)	Molds to the body shape in response to weight and heat, increasing contact area.	Low to moderate risk patients. Requires standard 2-hourly manual repositioning.
Active (Alternating Air)	Powered pump inflates and deflates alternate air cells in a timed cycle, mechanically shifting pressure points.	High risk patients or those with existing Stage 2+ injuries. Reduces, but does not eliminate, manual turns.
Fluid/Gel Immersion	Mimics the displacement of water to equalize pressure across the skin surface.	Often used in seating cushions for high-risk wheelchair users.

Heel Offloading Protocols

The heel bone (calcaneus) is extremely sharp and has almost zero subcutaneous fat padding. Heel ulcers deteriorate into bone infections rapidly.

- **Total Suspension:** The clinical goal is to "float" the heel. You should be able to pass your hand entirely under the patient's heel without touching it.
- **Pillow Method:** Place a standard pillow lengthwise under the calf (from knee to Achilles tendon). Do not place the pillow directly under the heel.
- **Boot Method:** Use commercial heel-suspension boots for restless patients. Ensure the boot is the correct size; too tight causes device-related injuries, too loose fails to float the heel.

Bed Mobility & Slide Sheets

Lateral bed repositioning requires sequential biomechanical steps to prevent shearing on the patient and back injuries for the staff.

1. **Bed Height & Flattening:** Lower the head of the bed flat to eliminate gravity shear. Adjust the bed height to the waist level of the shortest care worker to maintain safe ergonomics.

2. **Insert Slide Sheets:** Log-roll the patient to one side. Place two friction-reducing slide sheets directly on top of each other beneath the torso, hips, and shoulders. Roll the patient onto their back.
3. **Glide to Position:** With workers on both sides holding the top slide sheet close to the patient's body, perform a coordinated weight shift (front to back foot) to glide the patient into the new position.
4. **Remove Sheets:** Never pull the sheets aggressively against the skin. Remove the top sheet by sliding it against the bottom sheet, then remove the bottom sheet.

Skin Cleansing & Barriers

Routine washing with standard alkaline bar soaps destroys the skin's natural "acid mantle" (pH 4.5–5.5), increasing bacterial growth and dryness.

- **Cleansing:** Use pH-neutral or slightly acidic body washes. For incontinent patients, use specialized no-rinse foams that clean without excessive friction.
- **Drying:** Always pat the skin completely dry with a soft towel. Do not rub or scrub. Pay special attention to skin folds (under breasts, groin, abdomen) to prevent fungal moisture damage.
- **Barrier Creams:** For patients with incontinence, apply a medical-grade barrier cream (e.g., zinc oxide or dimethicone base) to protect the epidermis from caustic enzymes in urine and feces. Apply sparingly; thick layers trap heat.

Medical Device-Related Pressure Injuries (MDRPI)

Pressure injuries aren't just caused by beds and chairs. Any rigid medical device strapped to a patient can cause severe localized tissue necrosis.

Medical Device	High-Risk Anatomical Location	Prevention Strategy
Oxygen Tubing / Cannulas	Behind the ears, over the cheekbones, inside the nares.	Pad tubing with foam sleeves; check skin beneath the tubing every shift.
Urinary Catheters	Urethra, inner thigh, labia, or penis.	Secure the catheter tube firmly to the leg using a stat-lock to prevent dragging and tension.

Cervical Collars / Braces	Chin, occiput, clavicle, shoulders.	Ensure precise clinical fitting; inspect edges during daily hygiene routines.
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Special Care Populations

Bariatric Patients (Severe Obesity):

- **Atypical Locations:** Pressure injuries often occur inside deep skin folds (adipose tissue has poor vascularization), under the breasts, or beneath the abdominal apron (panniculus).
- **Equipment Limits:** Always verify the Safe Working Load (SWL) of the bed, mattress, and hoists. Standard mattresses will "bottom out" causing immediate injury.

Palliative / End-of-Life Patients:

- **Skin Failure:** In the final stages of life, multi-organ failure includes the skin (Kennedy Terminal Ulcers may appear suddenly).
- **Goal Shift:** The clinical goal shifts from strict prevention to comfort. If turning causes severe pain or distress, the physician may alter the care plan to prioritize pain management.

Clinical Documentation Rules

If a pressure injury develops or is discovered upon admission, immediate, objective documentation is legally required to establish a timeline of care.

Documentation Must Include:

1. **Exact Location:** Precise anatomical terminology.
 2. **Current Stage:** Stage 1, 2, 3, 4, Unstageable, or DTPI.
 3. **Measurements:** Length x Width x Depth in centimeters.
- **Upon Admission:** A full-body skin assessment must be completed on arrival. If a patient arrives with a Stage 3 ulcer and it is not documented, the facility assumes legal liability.
 - **Incident Reporting:** Any facility-acquired pressure injury (Stage 2 or above) must trigger a formal clinical incident report for root-cause analysis.
 - **Charting Turns:** Turn charts must be signed *at the time the turn occurs*, noting the exact position (e.g., "08:00 - Turned to Right 30° Lateral Tilt. Heels floated.").

Final Competency Checklists & Sign-Off

Use this final verification layout to confirm clinical performance benchmarks before signing off on independent pressure injury prevention duties.

Core Competency Indicator	Verified	Assessor Signature	Date
1. Identifies 5 major bony prominences at risk in supine/lateral positions.	☐		
2. Demonstrates accurate 30-degree lateral tilt repositioning using wedges.	☐		
3. Achieves total heel suspension utilizing the correct pillow or boot method.	☐		
4. Demonstrates safe slide sheet use for bed boosting without shearing.	☐		
5. Accurately differentiates a blanching vs. non-blanching Stage 1 injury.	☐		

Clinical Supervisor Declaration: I confirm the above-named practitioner has demonstrated practical competency in all core pressure injury prevention and repositioning requirements according to organizational WHS and clinical guidelines.

Supervisor Name: _____

Signature: _____

Date: _____